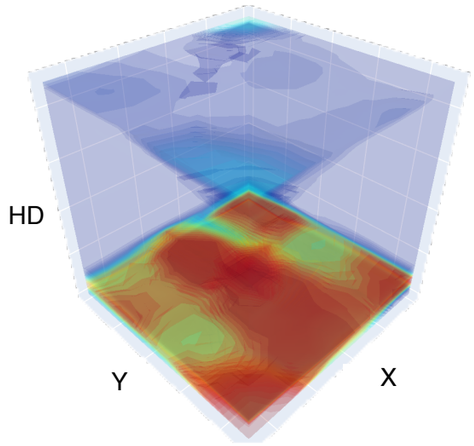
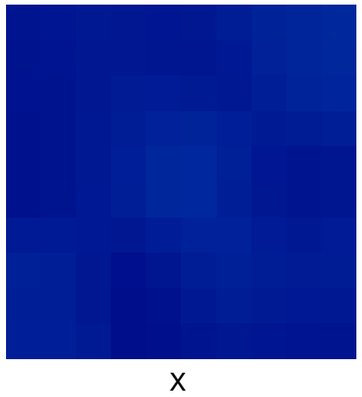
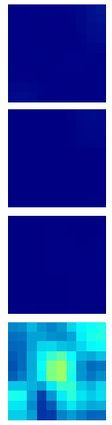
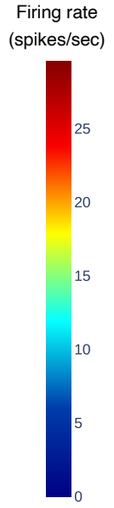
a

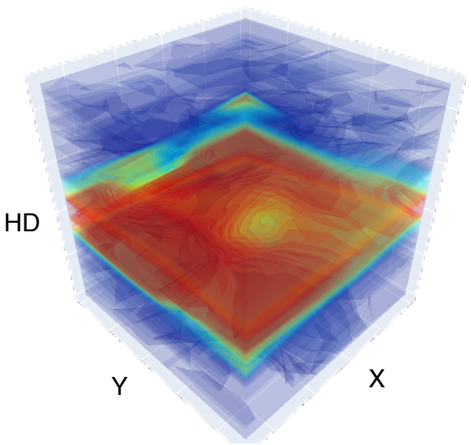

HD
Y
X

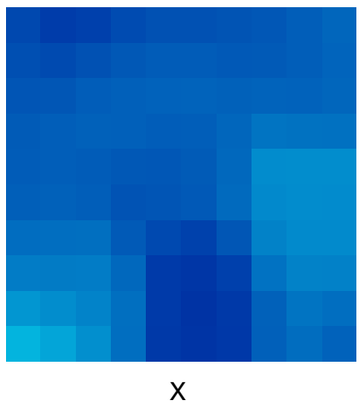

X

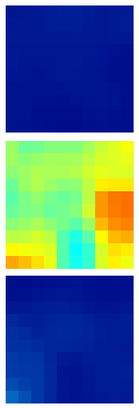


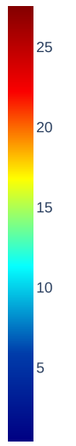

Firing rate
(spikes/sec)
25
20
15
10
5
0

b

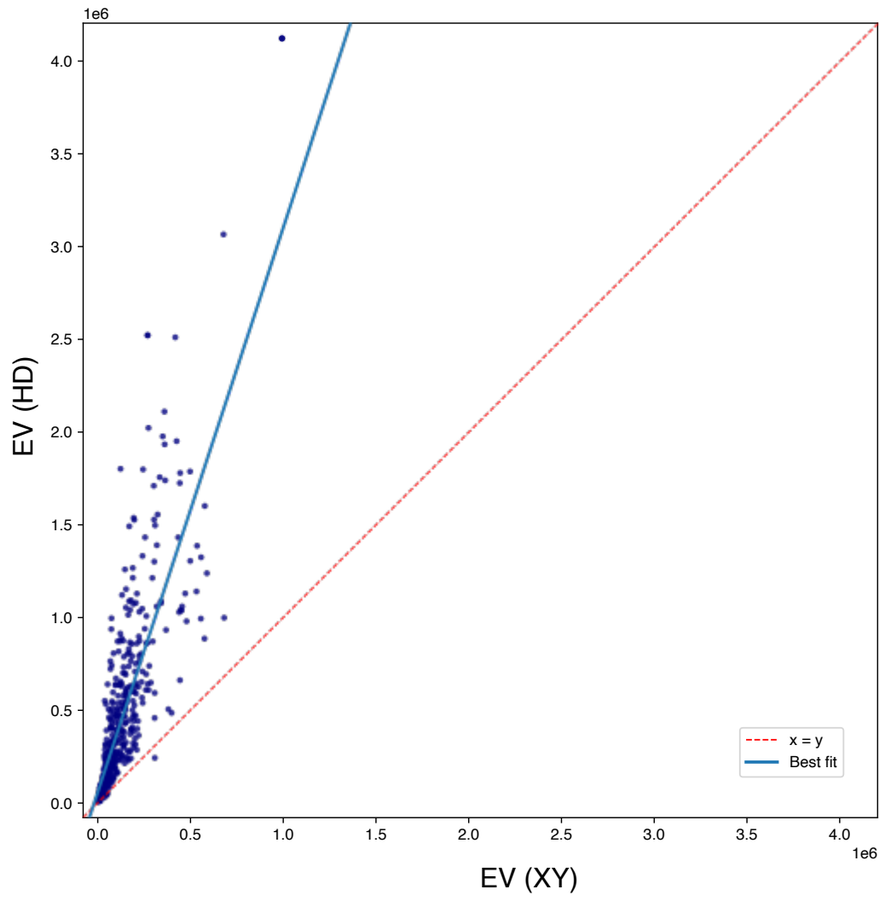

HD
Y
X


X

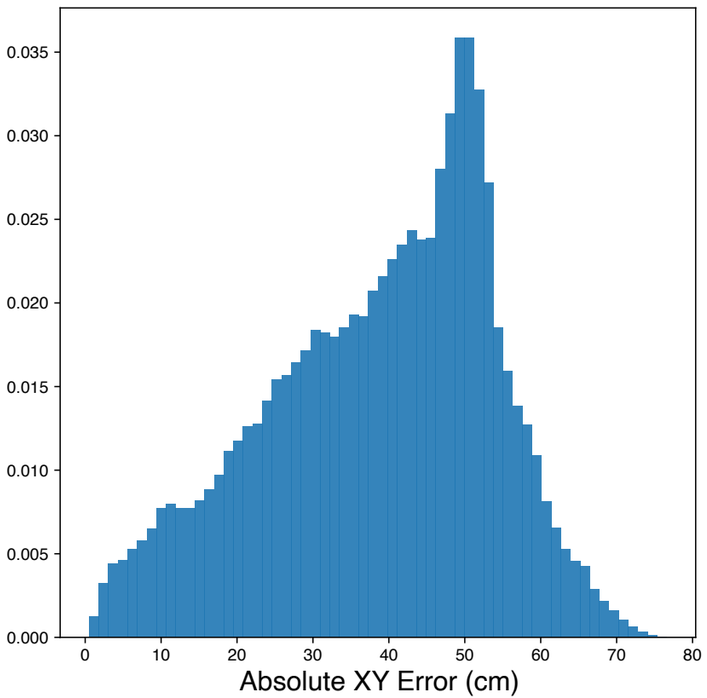


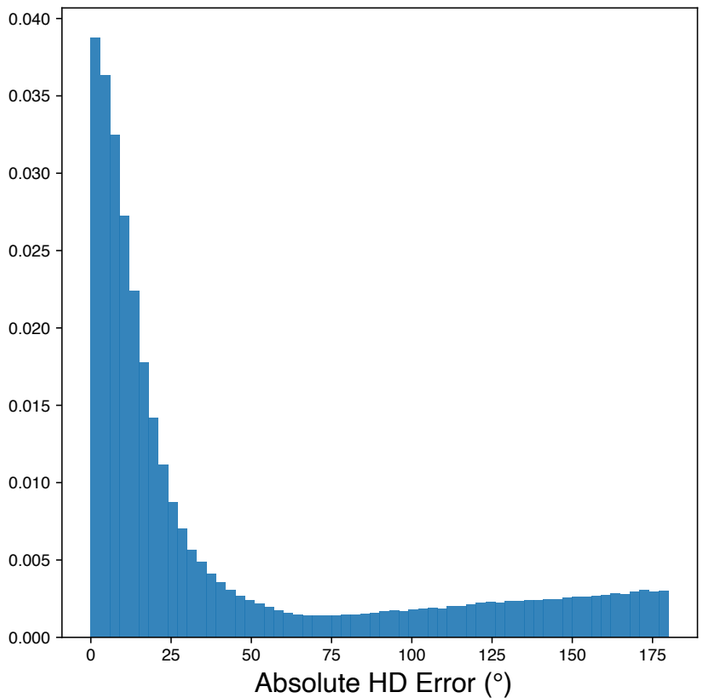

25
20
15
10
5

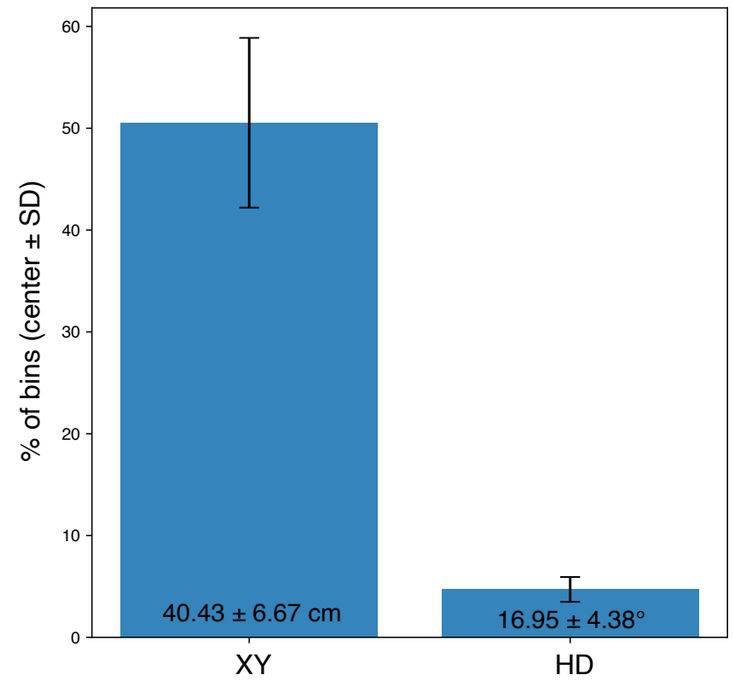
c


1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
EV (HD)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1e6
EV (XY)
x = y
Best fit

d


0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
10
20
30
40
50
60
70
80
Absolute XY Error (cm)


0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
25
50
75
100
125
150
175
Absolute HD Error (°)


60
50
40
30
20
10
0
% of bins (center ± SD)
40.43 ± 6.67 cm
16.95 ± 4.38°
XY
HD